Symmetrical Tree

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The problem

- Is the tree a mirror image of itself around its root?

Intuition

- A tree is symmetric if the left subtree mirrors the right.

Approach

- 1. Compare two nodes at a time
- 2. Values must match
- 3. Outer children pair together and inner children pair together

Java solution

```java
public class SymmetricalTree {
    public static boolean symmetricalTree(BinaryTree t) { return t == null || mirror(t.left, t.right); }

    private static boolean mirror(BinaryTree a, BinaryTree b) {
        if (a == null || b == null) return a == b;
        return a.value == b.value && mirror(a.left, b.right) && mirror(a.right, b.left);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Recap

- A tree is symmetric if the left subtree mirrors the right.
- Complexity: O(n)